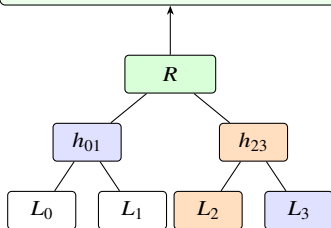


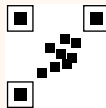
on-chain: `registerBatch( $R, n, cid, \dots$ )`



leaf: $L_i = H(H(i, \text{addr}(sk_i)))$, $i \in [0, n]$

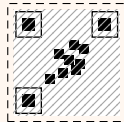
proof of unit 2: $\pi_2 = (L_3, h_{01})$, $\lceil \log_2 n \rceil$ hashes

Label of unit 2 (leaf L_2)



public QR

/u/c/b/2



secret QR

satk2:c:b:2:sk₂

secret layer is scratched off after purchase

Buyer app scans the secret QR and signs with the unit key:

$\sigma = \text{Sign}_{sk_2}(\text{EIP-712 ConsumeAuthorization}(b, 2, \text{consumer}, \text{deadline}, \text{chainId}))$

and sends it to `POST /v2/consume`; the relay attaches π_2 and pays the gas.

consume(req, π_2, σ) in SupplementRegistryV2:

- 1) $now \leq \text{deadline}$
- 2) `MerkleVerify( $\pi_2, R, L_2$ )` with L_2 rebuilt from (2, unitKey)
- 3) batch not recalled; the segment holding index 2 is at the point of sale; bit 2 of the bitmap unset
- 4) `ecrecover( $\sigma$ ) = unitKey = addr( $sk_2$ )`

$\Rightarrow$ set bit 2 and emit `UnitConsumed`; a second consume of the same unit reverts.